

Test 1 · Geometry

Grade 8

40 minutes 100 marks 11 questions No calculator

1–2	Revision of the Grade 7 course	Geometry 8, p. 3 · Stage 9 · 5.1
3–4	Polygons. Interior and exterior angles	Geometry 8, Tema 1 · Stage 9 · 5.2–5.3
5	Parallelogram and its properties	Geometry 8, Tema 2
6	Tests for a parallelogram	Geometry 8, Tema 3

- Draw a diagram for every question — an unlabelled answer earns no method marks.
- Give a reason for each step: the angle fact, the property, or the test you used.

What the paper weighs

TOPIC	MARKS	SHARE
Lessons 1–2 · Revision of the Grade 7 course	22	22%
Lessons 3–4 · Polygons. Interior and exterior angles	31	31%
Lesson 5 · Parallelogram and its properties	16	16%
Lesson 6 · Tests for a parallelogram	31	31%

Cancelling carries the largest share because every question in Section C also exercises the factorising from lessons 1–3.

A Section A · Six marks each

EASY 30 marks

One fact each. A short reason is enough.

1 Two angles of a triangle are 38° and 57° . Find the third angle. [6]

ANSWER

85°

WORKING

The angle sum of a triangle is 180° : $180^\circ - 38^\circ - 57^\circ = 85^\circ$.

2 An isosceles triangle has a base angle of 72° . Find its apex angle. [6]

ANSWER

36°

WORKING

Both base angles are 72° , so the apex is $180^\circ - 2 \times 72^\circ = 36^\circ$.

3 Each exterior angle of a regular polygon is 40° . How many sides has it? [6]

ANSWER

9 sides

WORKING

The exterior angles of any convex polygon add to 360° , so $n = 360^\circ \div 40^\circ = 9$.

- 4 In parallelogram $ABCD$, $\angle B = 115^\circ$. Find $\angle A$ and $\angle C$. [6]

ANSWER

$$\angle A = 65^\circ, \angle C = 115^\circ$$

WORKING

$\angle A$ and $\angle B$ are neighbouring, so they are supplementary: $180^\circ - 115^\circ = 65^\circ$. $\angle C$ is opposite $\angle B$, so it equals it.

Answering $\angle A = 115^\circ$ means opposite and neighbouring angles have been swapped.

- 5 In quadrilateral $ABCD$, $AB = CD$ and $AB \parallel CD$. Is $ABCD$ a parallelogram? Name the test you used. [6]

ANSWER

Yes — Test 1

WORKING

Test 1: one pair of opposite sides equal *and* parallel is enough.

B Section B · Ten marks each

MEDIUM

40 marks

Show the working. Method marks are available even when the final answer is wrong.

- 6 In $\triangle ABC$, $AB = AC$ and $\angle A = 40^\circ$. The bisector of $\angle B$ meets AC at D . Find $\angle BDC$. [10]

ANSWER

$$\angle BDC = 75^\circ$$

WORKING

The base angles are $(180^\circ - 40^\circ) \div 2 = 70^\circ$ each, so $\angle B = \angle C = 70^\circ$. The bisector halves $\angle B$, giving $\angle DBC = 35^\circ$. In $\triangle BDC$: $\angle BDC = 180^\circ - 35^\circ - 70^\circ = 75^\circ$.

- 7 The interior angles of a convex polygon add to 1440° . Find the number of sides, and the size of each interior angle if the polygon is regular. [10]

ANSWER

$$n = 10; \text{ each interior angle } 144^\circ$$

WORKING

$180^\circ(n - 2) = 1440^\circ$ gives $n - 2 = 8$, so $n = 10$. If regular, each angle is $1440^\circ \div 10 = 144^\circ$.

- 8 The perimeter of a parallelogram is 54 cm and one side is 5 cm shorter than the other. Find the two sides. [10]

ANSWER

$$16 \text{ cm and } 11 \text{ cm}$$

WORKING

Opposite sides are equal, so two adjacent sides add to 27 cm. With $x + (x - 5) = 27$: $2x = 32$, $x = 16$, and the other side is 11.

Dividing the perimeter by 4 is the usual error — the sides are not all equal.

- 9 In quadrilateral $ABCD$ the diagonals meet at O , with $AO = OC = 6$ cm, $BO = 5$ cm and $OD = 5$ cm. Is $ABCD$ a parallelogram? Give your reason. [10]

ANSWER

Yes — Test 3

WORKING

O is the midpoint of AC (since $AO = OC$) and of BD (since $BO = OD$), so the diagonals bisect each other. That is Test 3.

Compare with $AO = OC = 6$, $BO = 4$, $OD = 6$: only one diagonal is bisected, and the answer would be no.

C Section C · Fifteen marks each

HARD

30 marks

Set your work out as a proof: statement, then reason.

- 10 A convex polygon has interior angles adding to 1800° .
- (a) Find the number of sides. [5]
- (b) The polygon is regular. Find each interior angle and each exterior angle. [5]
- (c) Find the total number of diagonals. [5]

ANSWER

(a) $n = 12$ (b) interior 150° , exterior 30° (c) 54

WORKING

(a) $180^\circ(n - 2) = 1800^\circ$, so $n - 2 = 10$ and $n = 12$.

(b) Each interior angle is $1800^\circ \div 12 = 150^\circ$, and the exterior angle is $180^\circ - 150^\circ = 30^\circ$. Check: $12 \times 30^\circ = 360^\circ$.

(c) Each vertex sends a diagonal to $n - 3 = 9$ others, and every diagonal is counted from both ends:

$$\frac{12 \times 9}{2} = 54.$$

In (c) forgetting to halve gives 108 — the commonest slip in the topic.

- 11 $ABCD$ is a quadrilateral in which $AB = CD$ and $BC = AD$.
- (a) Prove that $ABCD$ is a parallelogram. [10]
- (b) A student says “ $AC = BD$ would also prove it”. Give a counter-example. [5]

ANSWER

(a) proof below (b) an isosceles trapezium

WORKING

(a) Draw the diagonal AC . In $\triangle ABC$ and $\triangle CDA$: $AB = CD$ (given), $BC = DA$ (given), and AC is common — so the triangles are congruent by **SSS**. Hence $\angle BAC = \angle DCA$, and these are alternate angles, so $AB \parallel CD$. Likewise $\angle BCA = \angle DAC$ gives $BC \parallel AD$. Both pairs of opposite sides are parallel, so $ABCD$ is a parallelogram.

(b) An isosceles trapezium has equal diagonals but only one pair of parallel sides, so equal diagonals are not sufficient.

Part (a) is Test 2, but the marks are for the congruence argument that proves it — quoting the test alone is not a proof. Part (b) is the discriminator.